Software Requirements Specification

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1. Introduction

1.1 Background

1.1.1 Software System Name

Intelligent curtain wall stone stain recognition system: The system aims to automatically identify and detect stains on the building facade through computer vision and deep learning technology. By providing real-time stain image upload, automatic detection, result display, report export and historical record query functions,

the system will improve the efficiency of building maintenance, reduce the cost of manual inspection, and help property management companies, building maintenance companies and other industry users manage the cleaning and maintenance of buildings more efficiently.

1.1.2 Project-related personnel

(1) Task Proposer:

This project was proposed by Mr. Huang Jie from the School of Computer Science and Software Engineering of Tongji University. It aims to solve the difficulties in the detection and cleaning management of stains on the exterior facade of buildings. It uses Computer Vision technology to provide an automated stain detection solution, reduce maintenance costs and improve management efficiency.

(2) Developer:

The development of this project is undertaken by students majoring in software engineering at Tongji University. The team consists of four members, Luo Shiyu, Bi Yalin, Fu Baoying and Wen Yujie. Under the guidance of the tutor, the team conducts project planning, requirement analysis, development, testing and deployment. The project team members have certain experience in computer vision, deep learning and software development, and can complete the development tasks with the help of the tutor and project guidance.

(3) User:

Building maintenance company: As the main user of the system, the company uses the system to automatically inspect the building facade and detect stains for timely cleaning and repair.

Property management company: Using this system for building cleaning management can quickly identify stains through the system and arrange maintenance personnel to handle them.

Architectural design company: Use the system to evaluate the quality of the building facade, check the impact of external pollution on the appearance of the building, and ensure the appearance quality.

1.1.3 Related organizations or systems

(1) User Management System :

The system will be integrated with the existing user management system to provide functions such as user authentication and permission management. The user management system will be responsible for managing user login, permission control, data access and other transactions to ensure that users with different roles can access the functions and data they need. The intelligent curtain wall stain recognition system interacts with the user management system through the API interface to complete identity authentication and permission assignment.

(2) Docker Backend Integration System :

The back - end service will be deployed in Docker containers to improve the flexibility, maintainability and scalability of the service. The Docker back - end integration system will manage the containerization of all back - end services to ensure the consistent operation of the system in different environments. The intelligent curtain wall stain recognition system will communicate with the Docker back - end system and call the services in the container for data processing, image recognition and result storage.

1.2. References

The following is the information cited and referenced in this specification:

1.2.1 Related technical documents

Title: YOLOv5: A Practical Object Detection Framework

Authors: Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi

Publication Date: 2020

Publication: IEEE Transactions on Pattern Analysis and Machine Intelligence

Source: IEEE Xplore database, for reference in the development of deep learning image recognition models.

1.2.2 Software Development Standards

Title: Software Engineering Standard - System Development Document

Document Number: GB/T 8567-2020

Publication Date: December 2020

Publisher: China Standard Press

Source: Standardization Administration of China.

1.2.3 Related Document Reference

Git Development Specification and Process for Backend Application Integration of Smart Curtain Wall Project (Released by Lin Jishen)

Database Usage Document (Released by Liu Yao)

Front - end Integration Specification (Released by Liu Yao)

1.3. Assumptions and Constraints

1.3.1 Funding Constraints

The project budget is limited. During the development process, technologies and tools with higher cost - effectiveness will be given priority, and over - investment in expensive hardware facilities will be avoided as much as possible.

1.3.2 Development Period

The project is expected to be completed between October 2024 and May 2025. The development cycle includes requirements analysis, system design, development and implementation, testing, optimization, and online deployment.

1.3.3 Equipment Conditions

Project development relies on existing computing devices, including high-performance development workstations and computing resources provided by the Cloud Computing Platform. Server resources come from the school or cloud computing service providers, and specific requirements are adjusted according to the project phase.

1.3.4 User Profile Preparation and Communication Issues

The user will provide the relevant building facade image data set and system requirements to ensure the smooth development of the project. The communication between the user and the development team needs to ensure clear requirements and timely feedback to avoid the lag of information communication.

1.3.5 Technical Feasibility

During the development process, it is assumed that the selected computer vision technology (such as YOLO model) can achieve the expected performance, and the computing platform provides sufficient computing power to support the training and inference of deep learning models.

1.3.6 System Compatibility

Assume that the system can be seamlessly integrated with the existing user management system and back - end infrastructure (such as Docker). The system will be designed to be highly scalable to facilitate future expansion of functions.

1.4. User Characteristics

1.4.1 Operator

(1) Educational level: The operators are building maintenance and property management company staff, usually with an education background above secondary level, and some may have college or undergraduate degrees.

(2) Technical expertise: Most operators' technical expertise is mainly concentrated in the field of building maintenance, and their computer technology capabilities are relatively basic. When designing the system, it is necessary to consider a simple user interface and easy-to-operate functions to ensure that non-technical personnel can quickly get started and use it.

(3) Expected frequency of use: The operator will frequently use the system to detect stains on the exterior facade of the building, and it is expected to operate multiple

times per day or week according to the building inspection plan.

1.4.2 Maintenance Personnel

(1) Educational level: The system maintenance personnel are professional technicians, usually with a high level of computer technology, and are able to handle technical problems and update and maintenance tasks that occur in the system.

(2) Technical expertise: The maintenance staff has certain system management experience and programming ability, and can solve system failures and carry out regular technical updates.

(3) Expected frequency of use: Maintenance personnel will regularly check and maintain the system to ensure its stable operation.

1.4.3 System Usage Frequency

Since the system is mainly used for the cleaning and maintenance management of building facades, the frequency of use is relatively high. Users will upload new image data every day or week, perform stain recognition tasks, report export tasks and view historical records.

2 Functional Requirements

2.1 System Scope

The main goal of the intelligent curtain wall stain recognition system is to help users detect stains on the building facade through Image Recognition Technology. The core functions of the system include: picture upload; stain recognition and result visualization; detection report export; historical record viewing.

2.1.1 Image Upload

(1) Function description: The user uploads pictures of the building facade through the system, supporting drag-and-drop upload/file selection upload and batch upload; the pictures need to be named after uploading; the uploaded pictures automatically enter the "to be detected" list; support deleting or modifying the uploaded pictures; select a photo from the "to be detected" list as the picture to be detected.

(2) User interaction: The system supports ordinary users to upload pictures one by one or in batches. After naming the pictures, users can select one picture from the picture list for detection, and can also add or delete pictures from the list.

(3) Requirements for uploading data: The pictures must comply with the specified format (such as JPEG, PNG, etc.) and size limit (not exceeding 50MB). The system will perform format verification, size limit, etc. on the uploaded pictures.

2.1.2 Stain Identification and Result Visualization

(1) Function description: When the user clicks the "Start Detection" button, the system uploads the picture to be detected to the back end; calls the back-end curtain wall segmentation and stain recognition model to segment the curtain wall blocks, identify the stain areas in the picture and mark the positions of the stains; returns the curtain wall block position marking map, curtain wall block segmentation map, curtain wall block stain marking map and curtain wall block stain marking map for result visualization display.

(2) User interaction: Ordinary users upload pictures through the interface. After clicking the "Start Detection" button, the system will automatically process them without additional operations.

(3) Processing flow:

Image upload: After the user clicks the "Start Detection" button, the system transfers the image to the backend for corresponding processing.

Curtain wall block recognition: The system uses the trained YOLO model to detect and label curtain wall blocks.

Curtain wall block segmentation: The system segments the original image into one or more curtain wall block images according to the curtain wall block annotation results.

Stain detection: Perform stain recognition on each segmented curtain wall block image and return the detection results, including the stain area and stain proportion.

Result visualization: The system returns the detection results to the front end for visualization.

2.1.3 Export Test Report

(1) Function description: On the stain detection page, users can export a report for the currently detected image; they can also export a report for a specific detection record on the history page.

(2) User interaction: After a user successfully detects a picture on the stain detection page, they can click the "Export Report" button to export the detection report; they can also click the "Export Report" button of a detection record on the history page to export the report.

2.1.4 View History

(1) Function description: Users can view all historical detection records, including information such as stain status, pollution level, curtain wall status, pictures uploaded each time, curtain wall marked pictures, curtain wall segmented pictures, identified stain areas, and detection time.

(2) User Interaction: Users can view past stain detection results on the query history page. The system supports filtering and querying by time.

(3) Data storage: All images and their detection results will be stored in the database, including metadata such as image paths, detection times, and stain locations, for easy access and viewing by users at any time.

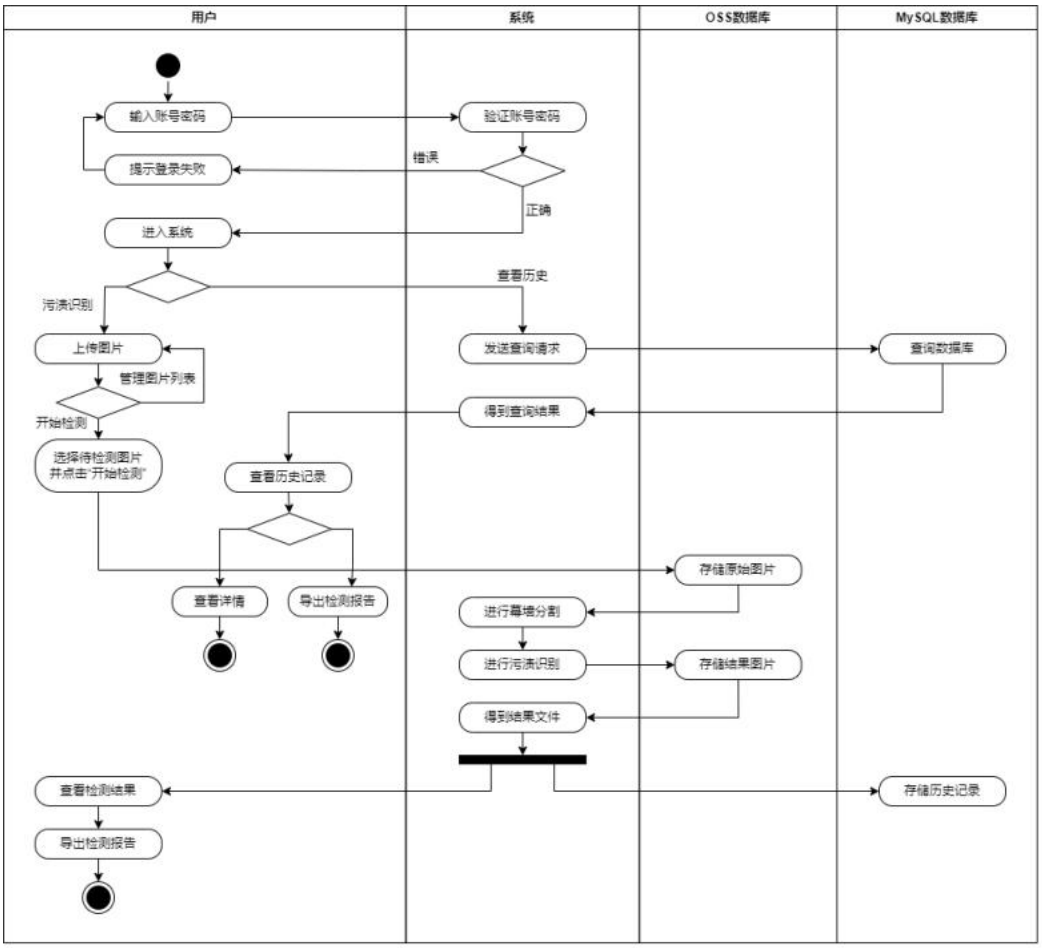
2.2. Overall System Process

The overall process of the system includes the complete process from the user uploading pictures to obtaining stain detection results and exporting reports, and then viewing historical records and exporting reports. The following is the overall activity diagram of the system:

2.3. Requirements Analysis Modeling

2.3.1. Functional Modeling

Functional modeling uses use case diagrams to represent the core functions of the system and user interactions. The following is a detailed description of the functional use cases and user personas:



- Use case 1: Upload a picture and identify the stain

Use Case	Upload Image
Use case number	UC001

Use case description	The user uploads one or more curtain wall images and enters the image name; the successfully uploaded images will automatically enter the "to be detected" list; the user can add or delete the images in the list; the user selects a single image from the list as the image to be detected.
Primary Actors	Individual users who need to detect the stain condition of curtain walls
Secondary Actor	system admin
Precondition	The user device supports uploading pictures, and the pictures meet the upload format.
Trigger Event	The user clicks the upload button or drags and drops to upload.
Postcondition	The system returns a list of images to be detected.
Main Event Flow	1. User uploads curtain wall image 2. User names the image 3. The system receives the picture data and temporarily saves it as a list to be detected. 4. The user adds or deletes the list to be detected 5. The user selects a single image to be detected

- Use case 2: Stain recognition and result visualization

Use Case Name	Stain recognition and result visualization
Use case number	UC002
Use case description	The user uploads the detected curtain wall image. Based on the YOLO model and image processing algorithm, the system segments and labels the curtain wall blocks, then detects the stain position and degree in the image for each curtain wall block image, and returns the processed image and analysis results.
Primary	Individual users who need to detect the stain condition of curtain

Actors	walls
Secondary Actor	system admin
Precondition	The user device supports uploading pictures, and the pictures meet the upload format.
Trigger Event	The user clicks the "Start Detection" button and submits the curtain wall picture.
Postcondition	The system returns the stain analysis results, including the stain level, the segmented image, and the labeled image.
Main Event Flow	1. The user clicks the "Start Detection" button and submits the curtain wall image to the backend 2. The system receives the picture data and saves it temporarily. 3. Curtain wall block recognition: The system uses the trained YOLO model to detect and label curtain wall blocks. 4. Curtain wall block segmentation: The system segments the original image into one or more curtain wall block images according to the curtain wall block annotation results. 5. Stain detection: Perform stain recognition on each segmented curtain wall block image and return the detection results, including the stain area and stain proportion. 6. The system returns the following information: the segmented curtain wall image; the pollution degree of the curtain wall with pollution; the original image and the marked image of the curtain wall with stains.

- Use case 3: Export test report

Use Case Name	Export inspection report
Use case number	UC003
Use case description	On the stain detection page, users can export a report for the currently detected image; they can also export a report for a certain detection record on the history page.

Primary Actors	Individual users who need to detect the stain condition of curtain walls
Secondary Actor	system admin
Precondition	The user has performed stain detection.
Trigger Event	The user clicks the "Export Report" button.
Postcondition	The system returns the stain detection report.
Main Event Flow	1. The user clicks the "Export Report" button to export the inspection report. 2. System download report.

- Use Case 4: View History

Use Case Name	User views historical detection records
Use case number	UC004
Use case description	Users can view their own detection records through the system, including stain status, pollution level, curtain wall status, pictures uploaded each time, curtain wall marked pictures, curtain wall segmented pictures, identified stain areas, detection time and other information.
Primary Actors	Individual users who need to detect the stain condition of curtain walls
Secondary Actor	system admin
Precondition	The system has saved the historical detection records for the user.
Trigger Event	The user clicks the "View Historical Detection Records" button to enter the record page.

Postcondition	1. The system shows the user an overview of the inspection records within the specified range. 2. The user clicks "View Details" to view the detailed results.

2.3.2. Data Modeling

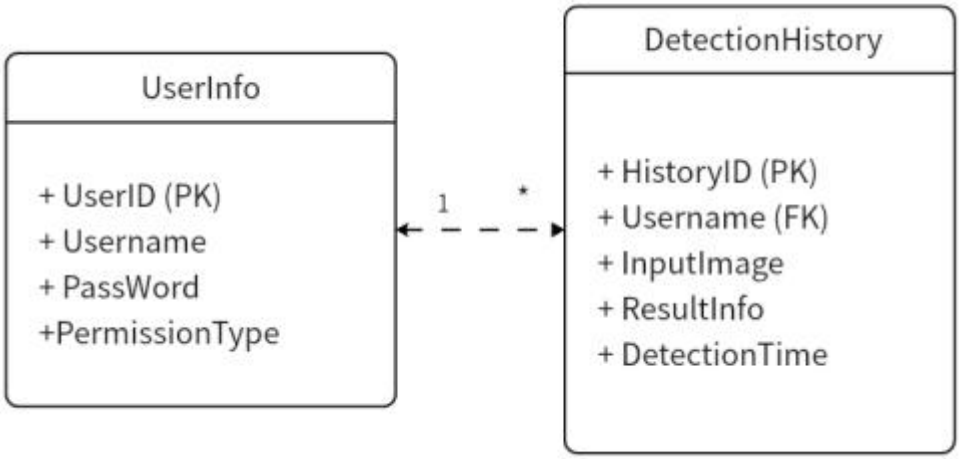
Data modeling uses entity relationship diagrams (ER diagrams) to represent the main entities involved in the system and their relationships. The main entities in the system are as follows:

(1) History Record

Attributes: History ID, Username, Uploaded Image, Recognition Result, Detection Time

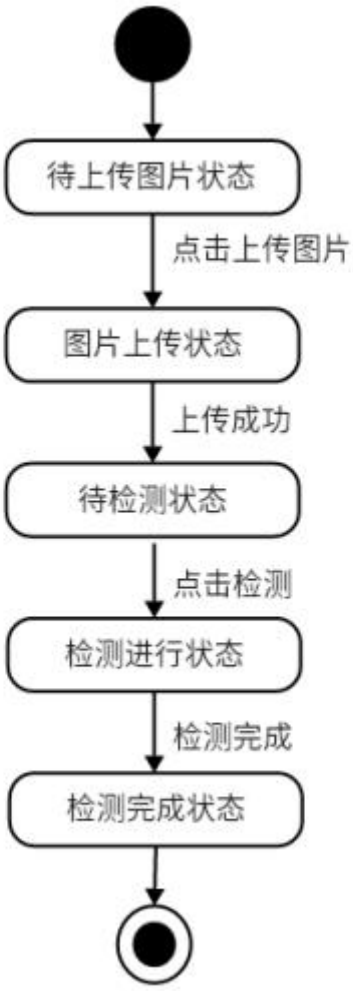
(2) User Information

Attributes: User Identification, User Name, Password, Permission Type



2.3.3. Behavior Modeling

(1) Upload pictures and stain detection status map



Status of the image to be uploaded:

The user has not uploaded a picture, and the system is waiting.

After the user clicks the upload button, the system enters the picture upload state.

Picture upload status:

After the user uploads the picture, the system starts to receive and store the picture file.

After the upload is successful, it enters the state to be detected.

Status to be detected:

In this state, the picture has not been detected yet.

After clicking Start Detection, the system enters the detection state.

Status is being detected:

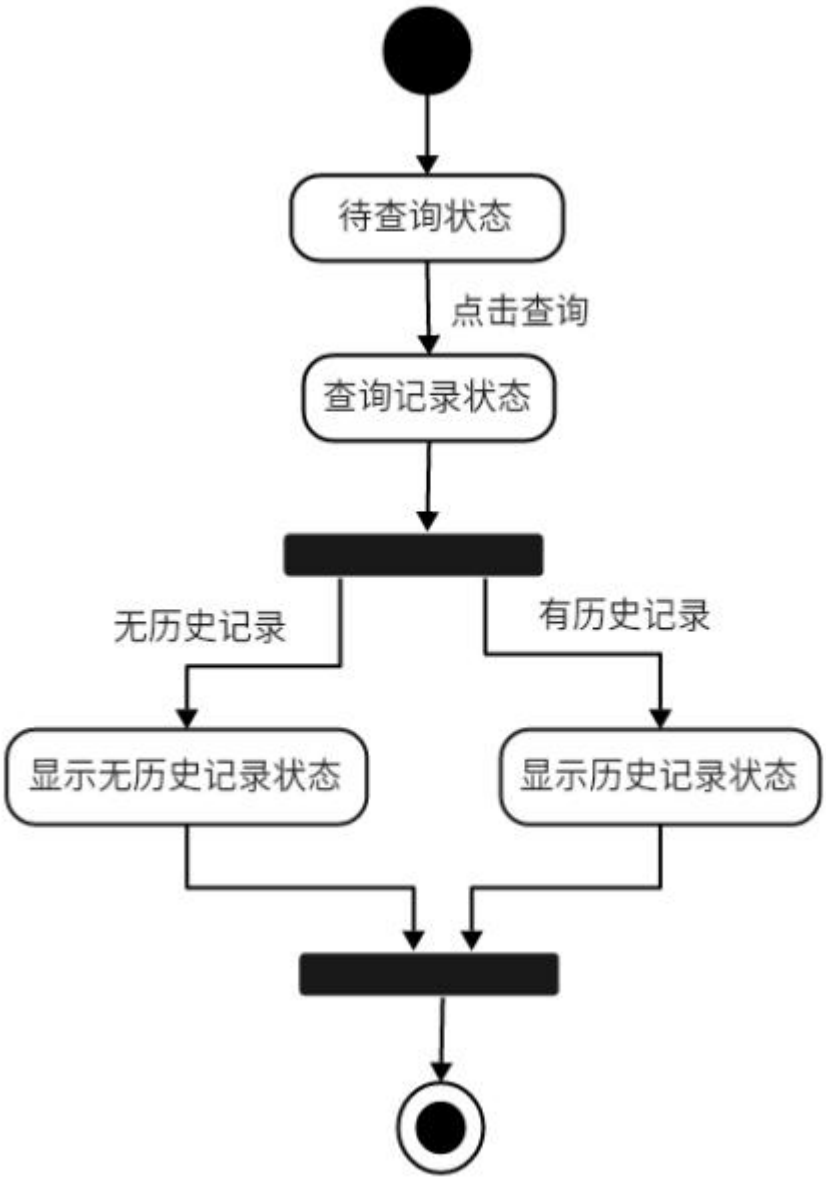
The system starts to analyze the uploaded pictures and identify stains.

The system processes the picture and identifies information such as the percentage of stains.

Detection completion status:

The system has completed the analysis of the picture and generated the detection results. The user can view the analysis results of the stain.

(2) View the history status diagram



Status to be queried:

Enter the historical record system and enter the state to be inquired.

After clicking the query, enter the query history state.

Query record status:

The system is querying the history record in this state.

After the query is completed, if there are records, it will enter the state of displaying historical records. If there are no records, it will enter the state of displaying no historical records.

Show the state of no historical records:

Prompt the user that there is no history.

Show history status:

Show the user's history.

3. Non-functional requirements

3.1. Performance Requirements

3.1.1. Accuracy

(1) Stain detection accuracy: The system should be able to accurately detect the stain area in the picture and give the percentage of the stain (accuracy requirement: more than 95%).

(2) Error range: When performing stain detection, the error range shall not exceed 5% to ensure that the results are close to the actual situation.

(3) Recognition of multiple stain types: The system should be able to recognize different types of stains (e.g., water stains, oil stains, etc.), and the recognition accuracy should be higher than 90%.

3.1.2. Time characteristics requirements

(1) Image upload time: After the user uploads the image, the system should complete the image reception within 8 seconds.

(2) Detection time: The system should be able to complete stain detection within 20 seconds to ensure a good User Experience.

(3) Historical record query time: When a user clicks to view historical records, the system should return the query results within 5 seconds.

(4) Response Time: The overall response time of the system (including upload, detection, and result display) should not exceed 50 seconds to ensure an efficient operational experience.

3.1.3. Input and Output Requirements

(1) Input Requirements: The system's input mainly consists of images uploaded by users, supporting common image formats such as JPEG and PNG. The image resolution should not exceed 4000x4000 pixels, and the file size should not exceed 10MB.

(2) Output requirements: The system should output the stain detection results, including stain pictures, stain percentage, stain location, etc. The historical record query results include the user's historical pictures and the detection results of each

picture.

3.2. Data management capabilities requirements

(1) Data storage: The system should support long-term storage of user data and historical detection results, and the storage time of historical records should be no less than 5 years.

(2) Data Update: Historical record data should be updated in real time. After the user uploads a new picture, the system should save the new data to the database within 5 seconds.

(3) Data query ability: The system should support efficient historical record query, support up to 100 users to perform query operations at the same time, and the query response time should not exceed 5 seconds.

(4) Backup and recovery: The system should have data backup and recovery functions, automatically backing up data once a day to ensure quick recovery in case of a failure.

3.3. Security and confidentiality requirements

(1) Data security: Encryption technology is used to encrypt and store the pictures uploaded by users and the detection results to ensure data security.

(2) Permission control: Users with different permissions (such as administrators and ordinary users) should have different operation permissions. Ordinary users can only view their own historical records, while administrators can view and manage the records of all users.

(3) Data transmission security: Data transmission must be encrypted through the HTTPS protocol to prevent it from being tampered with or stolen during the process.

(4) Privacy Protection: The system must comply with relevant privacy protection policies to ensure that users' personal information and uploaded images will not be leaked or misused.

3.4. Flexibility Requirements

(1) Scalability: The system should have good scalability and be able to easily support the addition of new functions, such as the possible addition of new stain types or more historical record display functions in the future.

(2) Compatibility: It should be able to be compatible with the operating environments of different platforms, such as desktop operating systems like Windows and Mac OS.

(3) Self-Adaptation: The system interface should have self-adaptation function, support display devices with different resolutions, and ensure the clarity and

operability of the interface.

3.5. Other special requirements

(1) User Experience: The system should provide a simple and intuitive User Interface with clear operation procedures to reduce the user's learning cost.

(2) Compatibility Requirements: The system should support common browsers for operation.

(3) Log management: The system should be able to record user operation logs, including operations such as uploading pictures and viewing historical records. The administrator can view the logs and conduct audits.

4. Operating Environment Requirements

4.1. Equipment

(1) Processor model and memory capacity

Recommended: Intel Core i5 or AMD Ryzen 5, with at least 8GB of RAM, 16GB recommended.

(2) External Storage Capacity

At least 500GB SSD storage space to ensure fast read and write.

(3) Input and Output Devices

Input device: Standard mouse and keyboard.

Output device: Monitor, at least 24 inches, resolution 1920x1080.

(4) Data communication equipment

Network device: Router supporting high-speed network.

4.2. Supported Software

(1) Operating System

Server operating system: Linux (such as Ubuntu or CentOS) is recommended.

Client Operating System: Windows, MacOS and Linux are supported.

(2) Database Platform

Database System: Use MySQL to store historical records; use OSS to store image data.

(3) Compile and test tools

Development Environment: PyCharm or VS Code.

Test tools: Postman, Apifox.

5. Appendix

5.1 Glossary

OSS (Object Storage Service)	Cloud storage service for storing pictures
MySQL Database	Stores system data, such as history, user information, etc.
Machine Learning model.	The deep learning model for stain recognition can analyze and identify the stain area in the picture
Permission Management	A mechanism in the system that defines and controls the permissions of different User Personas (such as administrators and regular users), ensuring that each user can only perform operations within their permissions.
Image URL	The URL of the image stored in OSS. The system accesses and displays the image through this URL.
Cloud Storage	Store data (such as pictures) in the cloud server, and users can access and manage it through the network.
User Experience (UX)	The overall User Experience of the user interacting with the system focuses on the convenience, smoothness and ease of operation, and improves the user's satisfaction with the system.
load balance	Ensure the system can maintain high performance and stability under high concurrency, and distribute the request traffic to multiple servers for processing.